FOR COMPLETE BEGINNERS

Wrong answer. No error.

Complete Modern Java, episode 5: use the debugger in Java and TypeScript

> Predict, pause, inspect, step, explain, fix. A debugger shows what your code does; you know what it should do.

The receipt prints Total: $4.99 with no error. It should be $3.99.

We stopped it halfway with a breakpoint, in Java and in TypeScript, and watched the wrong number appear.

In this episode

- Breakpoints: Run versus Debug
- Pausing before a line runs
- Step Over, Resume, Stop
- Free TypeScript debugging

Coming next

- Values and types (episode 6)
- Step Into (with methods, later)
- Tests (later in the course)

At the time of recording: Java 25, Node 24 LTS, IntelliJ IDEA 2026.1.3.

01 What the debugger showed

Run, with a breakpoint	`Ignored: Total: $4.99`
Debug, paused on the line	`bread 2, milk 1, breadPrice 1.5, milkPrice 0.99, no total`
Step Over	`total = 4.99`
Evaluate milk * milkPrice	`0.99`
Evaluate milk + milkPrice	`1.99`
After the fix, Step Over	`total = 3.99`

Paused on the total line, then one Step Over.

02 Java in IntelliJ IDEA

Click the line number for a breakpoint. Run ignores it; Debug stops on it, before the line runs.

Step Over runs one line. Resume runs to the next breakpoint or the end. Stop ends the session.

the bug

```
double total = bread * breadPrice + milk + milkPrice;   // 4.99
double total = bread * breadPrice + milk * milkPrice;   // 3.99
```

> Paused on the total line, Variables had no total row: the line had not run yet.
>
> Run on a Mac, 2026-10-23: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

03 TypeScript, free

The free IntelliJ IDEA has no TypeScript debugger. Node's inspector and Chrome DevTools are free.

terminal

```
node --inspect-brk groceryReceipt.ts
# Chrome: chrome://inspect, then "inspect" under Remote Target
# Sources: click line 6, Resume, read Scope, Step over
```

> DevTools showed the .ts file with each type annotation replaced by spaces: what Node runs.
>
> Run on a Mac, 2026-10-23: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

Debug it

1. Predict the answer first.
2. Set a breakpoint on the line that makes the value, and start with Debug.
3. Read the variables, then Step Over.
4. Explain the difference between your prediction and the value.
5. Challenge: find why the receipt says Items: 4.

Everything else

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The code and the challenge files: complete-modern-java
https://github.com/code-with-sam-dev/complete-modern-java

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